

Reproductive System

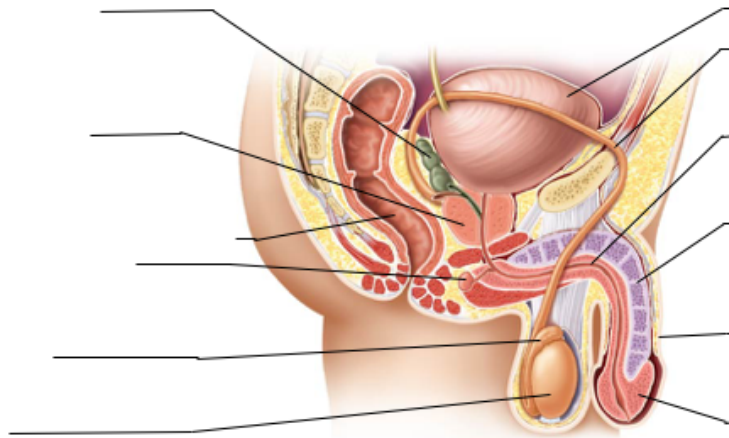
Today's Materials

- Fresh animal specimens (pig uterus and reproductive tract, testes, etc.)
- Mitosis and meiosis models
- Slides: Ovaries, testes
- Contraceptive Kit (borrowed from Foothill College Health Services Center)
- Colored pencils

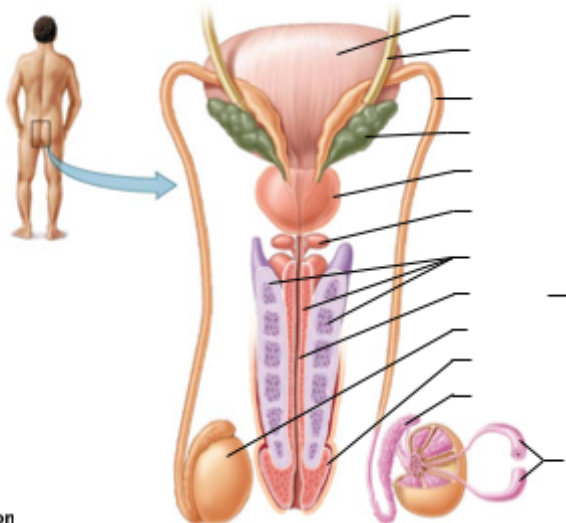
Activity 1: Gross Anatomy of the Human Reproductive System

Use your textbook to help you label the following diagrams. As you label the diagrams, match the pictures to the available models.

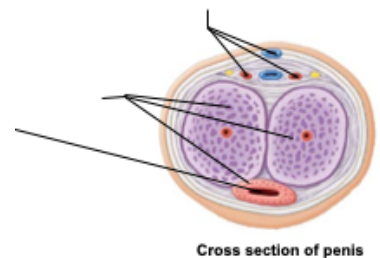
Male Reproductive System



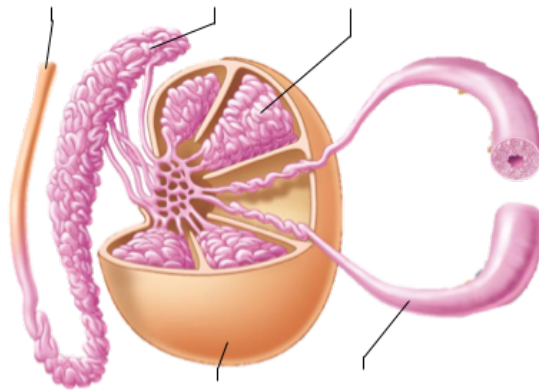
(a) Lateral view



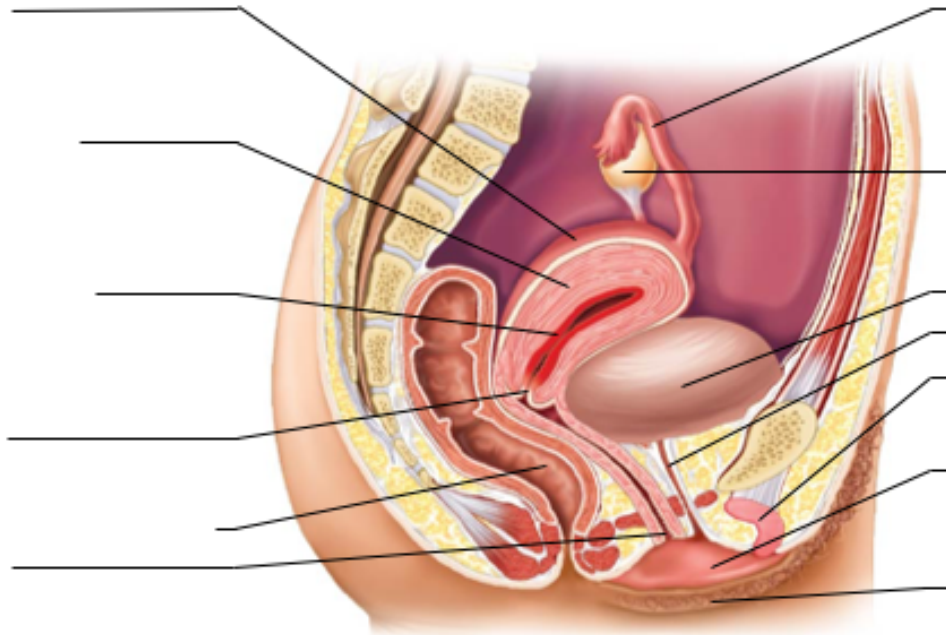
(b) Frontal Section



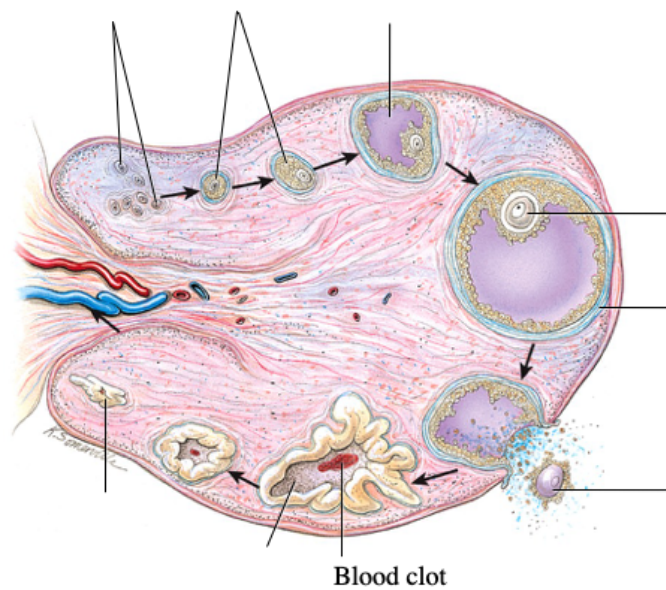
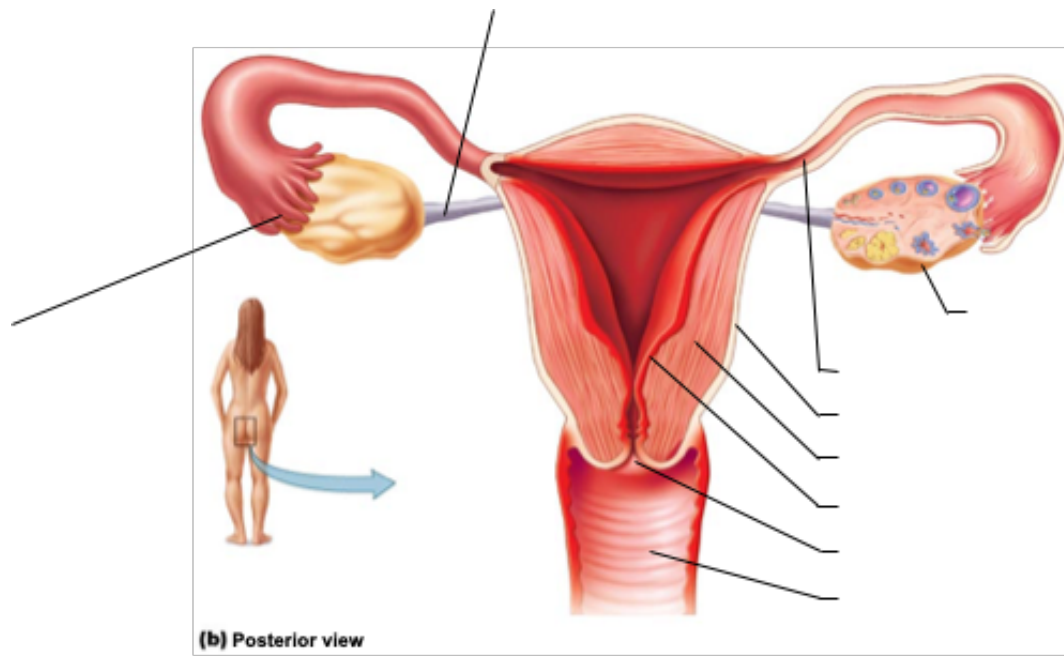
Cross section of penis



Female Reproductive System



(a) Lateral view



Activity 2: Comparative Anatomy in Animal Specimens

Observe the available animal reproductive system specimens. Identify as many of the structures from Activity 1 as you can. You can use the following online resources to help in your identification:

- Female Pig Reproductive Anatomy:
http://www.ansci.wisc.edu/jjp1/ansci_repro/lab/female_anatomy/pig_2010/pig_tract_display_initial.htm
- Bull Reproductive Anatomy:
http://www.ansci.wisc.edu/jjp1/ansci_repro/lab/lab2/bull_tract/bull_tract_gen.html

Place a pin in the specimen marking the structure.

1. In what ways do these structures differ from that of the human?
2. How are the structures similar?
3. What was most surprising to you?

Activity 2: Cellular Reproduction -Mitosis and Meiosis

Mitosis is the process by which the body produces two genetically **identical** daughter cells. In humans, mitosis functions to make new cells for growth and development and for tissue replacement. A second type of cell division is **Meiosis**, the process by which we produce gametes – eggs and sperm-which consist of half of the genetic material of the parent cell.

Review and compare the steps of mitosis and meiosis.

Mitosis

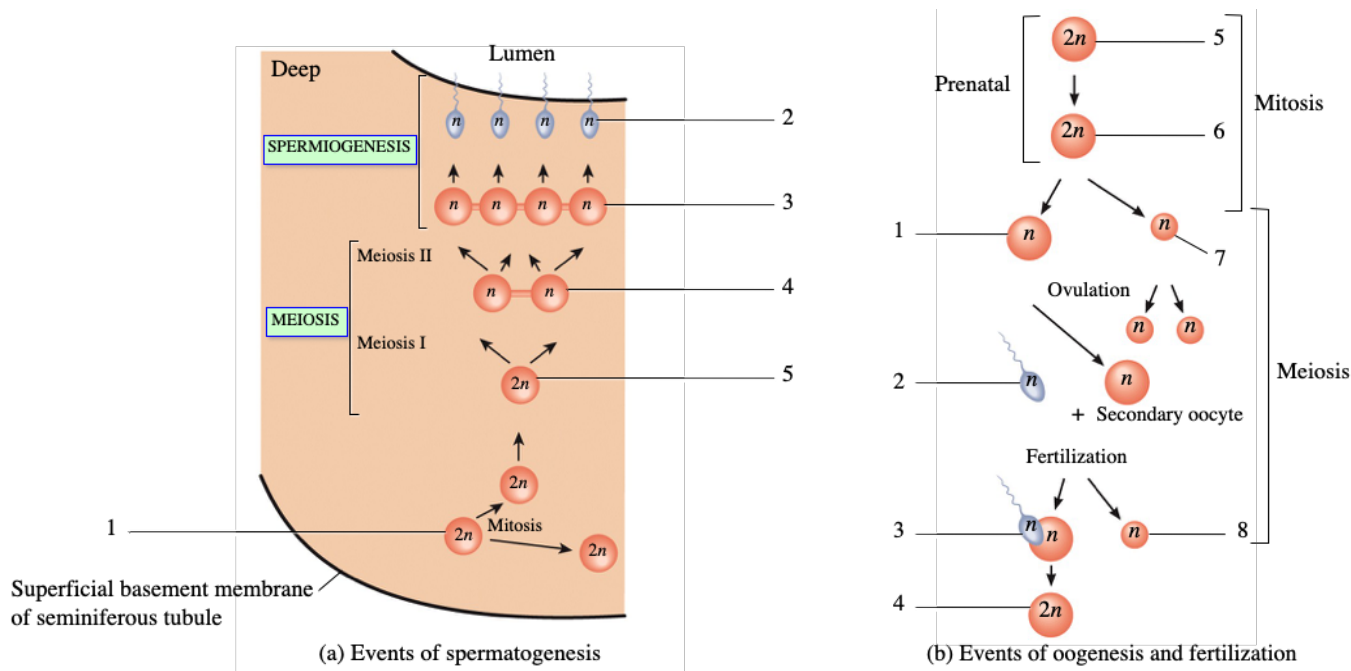
1. Shuffle and arrange the cell mitosis models to place them in order. (Alternatively, your instructor may provide you with mitosis cards to place in order).

Meiosis

2. Shuffle and arrange the cell meiosis models to place them in order. (Alternatively, your instructor may provide you with meiosis cards to place in order).

Compare

3. Make a table in your notebook comparing and contrasting the processes of mitosis and meiosis. Some aspects to consider include: rounds of cytokinesis, resulting ploidy of cell nuclei, arrangement of chromosomes during metaphase, etc.
4. The diagrams below depict the events of spermatogenesis, oogenesis and fertilization. Label the structures indicated with numbers. As you do so, keep in mind the differences between mitosis and meiosis.

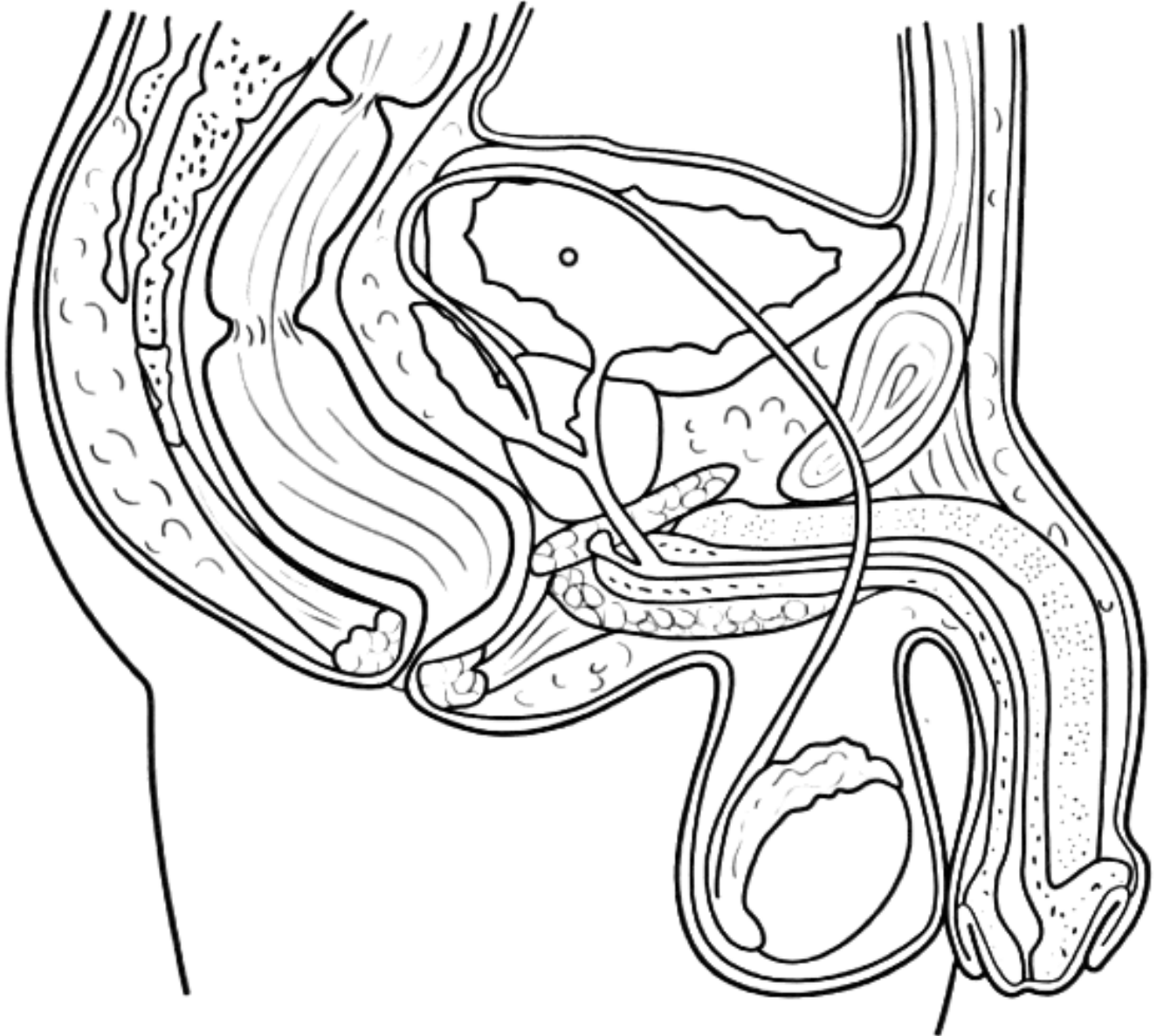


Activity 3: Microscopic Anatomy of Ovaries and Testes

- View a slide of testis draw and identify:
 - Seminiferous tubules**
 - Leydig (interstitial) cells**
 - Spermatogenic cells**
 - *You are not expected to distinguish among cell types in various phases of spermatogenesis. The term "spermatogenic cells" refers to spermatocytes and spermatids, collectively.
- View a slide of the Uterus. Draw and identify the following:
 - Myometrium**
 - Endometrium**
 - Endometrial glands**
- View a slide of the human Ovary. Draw and identify the following:
 - Primordial follicle**
 - Primary follicle**
 - Graafian (mature) follicle**

Activity 4: Tracing the Journey of Sperm

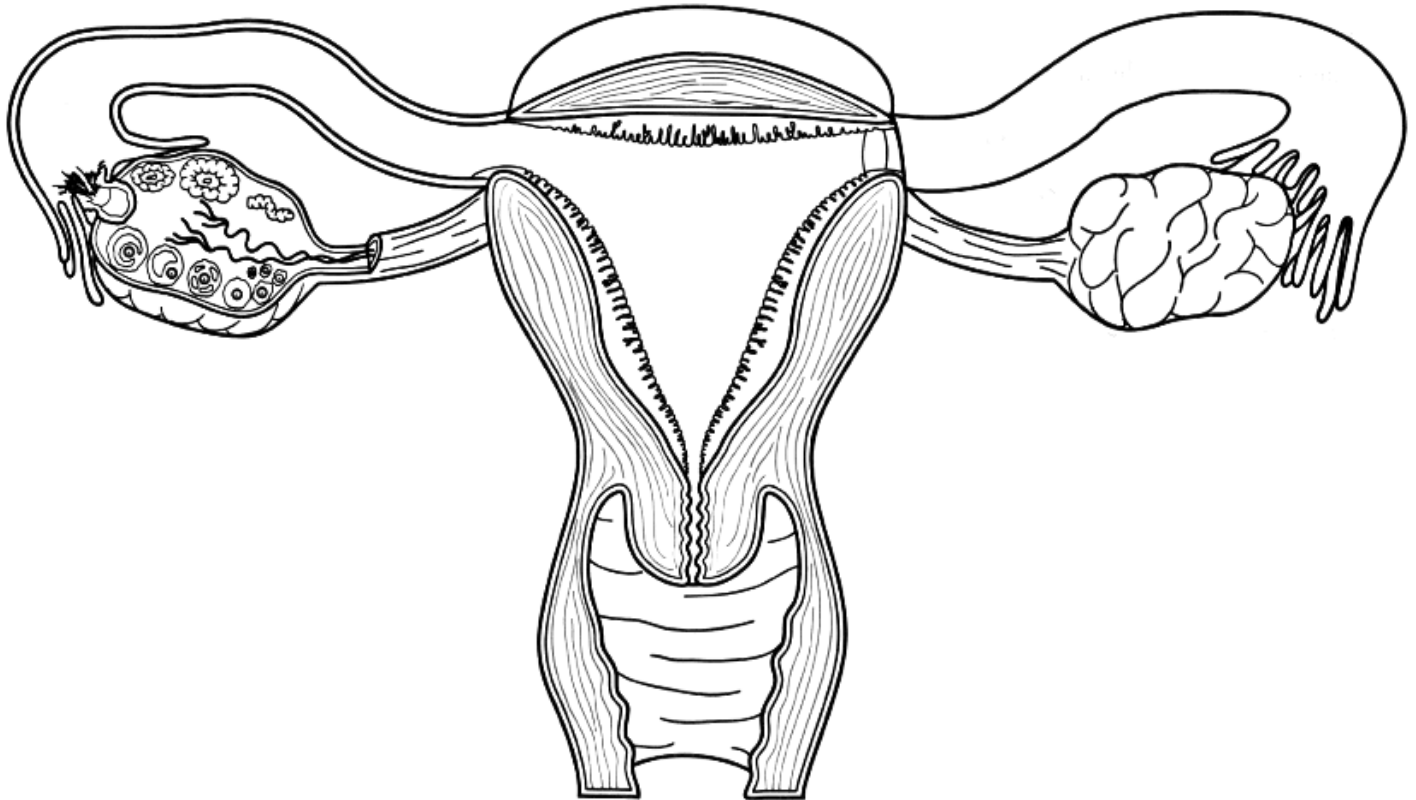
Draw arrows on the diagrams below to trace the journey of sperm from the site of spermatogenesis in the testes to the point of fertilization in the female reproductive tract. List the organs and structures that the sperm travels through below the diagrams.



List the organs the sperm travels through from the testes until it leaves the male reproductive tract:

Draw arrows on the diagram below to trace the journey of the sperm as it enters the female reproductive tract until it reaches the egg.

Next, use a contrasting color to trace the journey of the fertilized egg from fertilization to implantation in the uterine wall.



List the organs the sperm travels through on its journey through the female reproductive tract:

List the organs the fertilized egg travels through on its journey to implantation:

Activity 5: Contraceptive Methods

Observe the available contraceptive devices found in the Contraceptive Kit. Note that there are a variety of devices including barrier methods, hormonal methods, spermicidal methods, and even methods that prevent embryo implantation.

- Take a look at each contraceptive device (you may need to read the instructions), determine how the device prevents pregnancy, and locate the point in the human reproductive tract where it stops the process of reproduction (it may be in the male or female system!).
- Use contrasting colored pencils to indicate on the diagrams in activity 4 where the sperm, egg, or zygote would be stopped.

Contraceptive Device	How it prevents pregnancy.	Location in the human reproductive tract where it takes effect.	Advantages	Disadvantages	Potential Side Effects

** If you have questions or concerns about contraception or prevention of sexually transmitted diseases, visit the Health Services in room 2126 (near the Bookstore) or call (650) 949-7243.

1. Imagine you are a physician working with a patient. What might you consider when advising your patient on contraceptive options?
2. Read the following case-studies and with your group, decide which birth control method would be the best choice for each patient. Explain your reasoning!
 - a. Stacy is in a mutually monogamous relationship. Because of an underlying medical condition, she cannot use any hormonal birth control method. She is looking for a method that will protect her from pregnancy for the next 5 years, minimum. What would be a good method for her?

- b. Leslie is breastfeeding her newborn boy, and she needs a method of birth control. Protection from sexually transmitted infections is not a high priority, and she and her husband do not want to use condoms. What would be a good method for her?
*Consider: Estrogen and progesterone hormones can be secreted in breast milk.
- c. Greg is a 43 year old recently single male who definitely does not want to have children. He is not currently in a monogamous relationship, but would like to start dating. What birth control method would you suggest for Greg?
- d. Monique is in a long-term relationship needs a reliable method of birth control. She was taking "the pill" for a while, but she had a difficult time remembering to take a pill every day. She thinks she may want to have a child in about a year, so she doesn't want a method that might delay fertility. What would be a good method for her?
- e. Leticia and Robert are married. Leticia has completed her childbearing and the couple is looking for a long-term, foolproof birth control method. What would be a good method for them?